Fixed-label oriented CM lifts

Exact finite reduction · P-192 primary; five-curve certificate replay

2026-11-08 · T-1888 · PR 3674

Derived search object

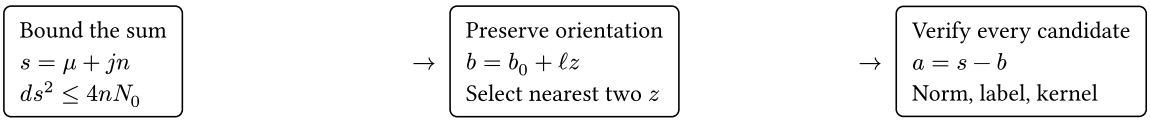
For an ordinary prime-field curve, work in the Frobenius order $\mathbb{Z}[\pi]$, with $\pi^2 - t\pi + p = 0$, $n = p + 1 - t > 0$ and $d = 4p - t^2 > 0$. Fix the existing rational-subgroup label μ and selected signed Frobenius eigenline. For $\alpha = a + b\pi$, put $s = a + b$. Substitution and completion of the square give

$$N(\alpha) = s^2 + (t-2)sb + nb^2,$$

$$4nN(\alpha) = (2nb + (t-2)s)^2 + ds^2.$$

Hence every lift satisfies $N(\alpha) \geq d\frac{s^2}{4n}$, with $s = \mu + jn$. This is an exact algebraic identity and bound, without a class-number estimate.

Finite search for the same scalar label and selected eigenline



Derived algorithm, not an executed minimum search. Enumeration includes every sum below the initial norm bound. Evaluation cost is a separate obligation.

Orientation and nearest coefficients

Let the distinct roots modulo the odd prime $\ell \neq p$ be ρ, ρ' and assume $\rho \neq 1$. The chosen kernel is annihilated exactly when

$$s + (\rho - 1)b \equiv 0 \pmod{\ell}.$$

Set $b_0 \equiv -\frac{s}{\rho-1} \pmod{\ell}$. To retain the opposite eigenline require $b_0 \neq 0$; if $b_0 = 0$, omit this sum. All remaining coefficients have $b = b_0 + \ell z$. At fixed s, only the square changes, so the minimum occurs at the one or two nearest integers to

$$z_* = \frac{-(t-2)s - 2nb_0}{2n\ell}.$$

Retain both on a tie and set $a = s - b$. The opposite eigenvalue is $b(\rho' - \rho)$, which is nonzero modulo ℓ. The exceptional $\rho = 1$ case needs a separate branch; all five current roots differ from 1.

A complete finite interval

Given any existing oriented lift of norm $N_0 > 0$, every candidate of norm at most N_0 satisfies

$$|s| \leq B, \quad B = \left\lfloor \sqrt{\left\lfloor \frac{4nN_0}{d} \right\rfloor} \right\rfloor.$$

Enumerate only $s = \mu + jn$ in this interval and the nearest coefficients above. The old candidate is feasible, so this covers a global norm minimum within the specified order and orientation. A replay certificate must include and verify the interval's square inequalities, both integer endpoints, every admitted sum and nearest pair, the winner and all label/kernel checks. There is no heuristic omitted tail.

What the point action establishes

The global endomorphism has the straight-line action $\alpha(P) = [a]P + [b]\pi(P)$. On the rational subgroup it is exactly $[s]P$, with scalar label $s \bmod n = \mu$. Charge the group operations actually used. Changing the CM lift's norm does not by itself change this scalar action or provide a runtime improvement.

Annihilating the certified kernel gives a factorization $\alpha = \psi\varphi$ through its separable quotient. The residual degree is $\frac{N(\alpha)}{\ell}$. This existence statement supplies neither expanded quotient polynomials nor a cheaper point evaluator for ψ.

The existing explicit closed dual route still proves $\sigma\eta\varphi = [\ell]$ by full polynomial identity and differential sign. The charged rational-subgroup correction $\left[\frac{\mu}{\ell}\right]$ reaches the desired label; it is not thereby identified with the full global quotient. A new route must retain its endpoint isomorphism and full point action separately from its scalar label.

Qualification and scope

The native GMP integer replay independently checks the five existing P-192, P-224, P-256, P-384 and P-521 certificates. It recomputes trace, norm, completed square, rational label, selected/opposite eigenline and quotient degree. Fifteen altered norm/label/orientation controls are rejected; final orientation controls substitute the other valid Frobenius root. This is software qualification against published inputs, not an executed minimization, benchmark or formal proof.

The first Boost-based compile failed because its headers were absent; its source and failure are preserved. The corrected implementation uses GMP 6.2.1 and retains dependency/build/check receipts. Invocations have a 120-second ceiling, one process/thread and 2 GiB address-space ceiling. No cloud job or cryptographic clock was launched. The separate modular-recoding report retains the previously frozen 30,780-row count census; it is unchanged by this derivation.

The reduction covers $\mathbb{Z}[\pi]$. A larger endomorphism order may have elements with denominators in this basis; extending the bound needs a certified ring index. The admissible set fixes one eigenline at ℓ; separability and horizontal factorization at other primes remain route checks. Adding restrictions needs a new completeness argument. No minimal-norm five-curve result or efficient quotient is claimed here. Future comparative execution must publish and freeze its source, inputs, finite resources, tie rule and full evaluation costs before data.

Sources and next construction

The standard degree quadratic form and rational Frobenius fixed-point identity are checked in Sutherland's MIT 18.783 Fall 2023 Lecture 7, pages 1-2. Endomorphism operations and dual cancellation are in Lecture 6, sections 6.2-6.3. Both primary PDFs were read; the completed-square reduction is derived above. No novelty claim is made.

The next construction is a complete integer certificate enumerating this finite interval for the unchanged labels and orientations, followed by a costed executable quotient or other closed route. The current finite-bound derivation is complete at its algebraic scope; neither that next search nor its runtime evaluation has been performed.

Primary sources: math.mit.edu/classes/18.783/2023/LectureNotes6.pdf and LectureNotes7.pdf. Full derivation and access gaps: ../../CM_LIFT.md. Inputs: ../../fixtures-v2/published-curves.json and marked-relations.json. Replay: ../../lift_identity.cpp and ../../qualification/lift-identity-v3/. KB unavailable: no loaded MCP tools; migrated scripts/research-kb wrapper absent, exit 127. No successful prior-art absence claim or new ecdb query is inferred.